

Unfolding the Drell-Yan Absolute Cross-Section: Methodology and Final Results

Chatura Kuruppu

New Mexico State University
SeaQuest Experiment (E906)

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- Events from runs 2 and 3 (Input Files Used)
- Event Selection Criteria (Chuck Cuts)
- POT values Used
- Theory: Iterative Bayesian Unfolding
- Step-by-Step Procedure
- Unfolded Single Differential Cross-Section as a function of p_T
- Multi-Dimensional unfolding for x_F and Invariant Mass phase space

Kinematic Phase Space:

```
MASS_BINS = [3.9, 4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.8, 10.0]
PT_BINS   = [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8, 2.5]
XF_BINS   = [-0.05, 0.00, 0.05, 0.10, ..., 0.80, 0.85]
```

Events from runs 2 and 3 (Input Files Used)

Note: Currently using all the runs saved in runs 2 and 3.

Harsha's ROOT files:

/sequest/users/harshaka/e906.project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/:

- merged_RS57_LH2.1.1138.root
- merged_RS57_LD2.3.1138.root
- merged_RS57_Empty.2.1138.root
- merged_RS59_LH2.1.465.root
- merged_RS59_LD2.3.466.root
- merged_RS59_Empty.2.466.root
- merged_RS62_LH2.1.1234.root
- merged_RS62_LD2.3.1237.root
- merged_RS62_Empty.2.1234.root
- merged_RS70_LH2.1.264.root
- merged_RS70_LD2.3.266.root
- merged_RS70_Empty.2.267.root

Abi's ROOT files:

/sequest/users/apun/e906.projects/rs67_merged_files/:

- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root

Event Selection Criteria (Chuck Cuts)

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos \theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

Track Quality & Acceptance:

- $\chi^2_{\text{target}} < 15$, $\chi^2_{\text{dimuon}} < 18$, and $\chi^2/(n\text{Hits} - 5) < 12$.
- Hit requirements: $n\text{Hits} > 13$ per track, $n\text{Hits}_1 + n\text{Hits}_2 > 29$, St1 total > 8 .
- Track vertex longitudinal limits: $-320 < z_v < -5$.
- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for $\text{runID} \geq 11000$ and 0.4 otherwise.

Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.
- $20 < D1 < 385$ (global reconstruction efficiency curve defined in this range)

ONLY RS62, $\text{runID} > 11500$ applied to ensure correct target positions of dimuons

Note: Currently using the same set of Chuck Cuts defined in DocDB 2111-V42

POT values Used

Roadset	LH2	LD2	Flask
57	3.533324×10^{16}	1.768358×10^{16}	3.918550×10^{15}
59	9.365986×10^{15}	4.319952×10^{15}	1.010350×10^{15}
62	5.281659×10^{16}	2.382899×10^{16}	1.097456×10^{16}
67	1.611435×10^{17}	7.694541×10^{16}	3.662417×10^{16}
70	1.785745×10^{16}	8.752588×10^{15}	3.841280×10^{15}

Note: POT Values from runs mixed by Harsha in DocDB 11524.

Theory: Iterative Bayesian Unfolding

We utilize the **D'Agostini Iterative Bayesian** approach implemented in the RooUnfold framework.

- **Response Matrix (R_{ij}):** Encapsulates detector effects (smearing/efficiency), mapping True (T_j) to Measured (M_i).
- **Bayes' Theorem:** Reverses the mapping to find the probability that an event in measured bin i came from true bin j :

$$P(T_j|M_i) = \frac{P(M_i|T_j)P_0(T_j)}{\sum_k P(M_i|T_k)P_0(T_k)}$$

- **Iterative Update:** The unfolded yield in bin j is estimated, and then used as the prior for the next iteration:

$$\hat{n}(T_j) = \frac{1}{\epsilon_j} \sum_i P(T_j|M_i)n(M_i)$$

Step-By-Step Procedure

The multidimensional unfolding process follows these logic steps:

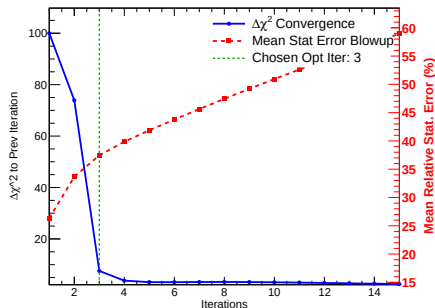
- ➊ **Yield Extraction:** Subtract background (Flask) from LH2 and LD2 target yields.
- ➋ **Response Training:** Generate a Target-Specific Response Matrix (Mass vs x_F) using high-statistics Monte Carlo.
- ➌ **Unfolding:** Apply RooUnfoldBayes to the subtracted yields across the 2D phase space.
- ➍ **Optimization:** Determine optimal iterations per bin by balancing $\Delta\chi^2$ convergence against Statistical Error blow-up.
- ➎ **Acceptance Correction:** Apply acceptance corrections to the unfolded results.
- ➏ **Cross-Section Calculation:** Normalize by luminosity and nucleon properties:

$$\frac{d^2\sigma}{dMdx_F} = \frac{N_{unfolding}}{\mathcal{L} \cdot \Delta M \cdot \Delta x_F \cdot A}$$

Unfolding Optimization

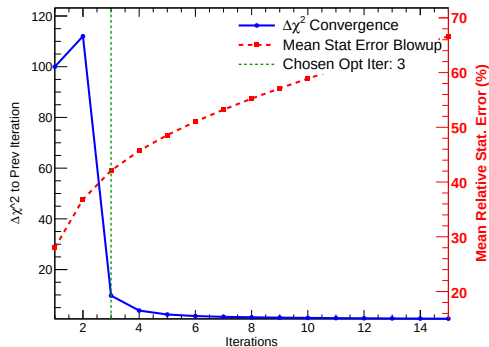
LH2 (Messy): $0.70 \leq x_F < 0.75$

Unfold Optimization LH2 Messy xF Bin 15



LD2 (Messy): $0.70 \leq x_F < 0.75$

Unfold Optimization LD2 Messy xF Bin 15



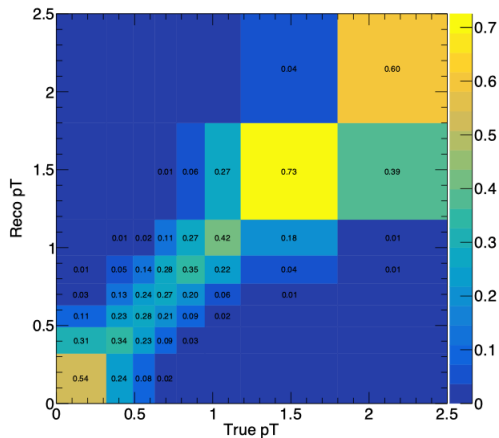
Mean Relative Statistical Error

$$\text{Mean Stat. Error (\%)} = \frac{1}{N_{\text{bins}}} \sum_{i=1}^{N_{\text{bins}}} \left(\frac{\sigma_i}{N_i} \right) \times 100\%$$

Response Matrices (clean)

LH2 Target

LH2_clean Normalized Response: pT



LD2 Target

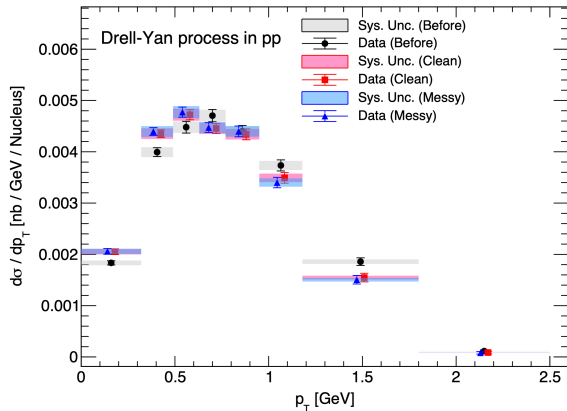
Response Matrices (messy)

LH2 Target

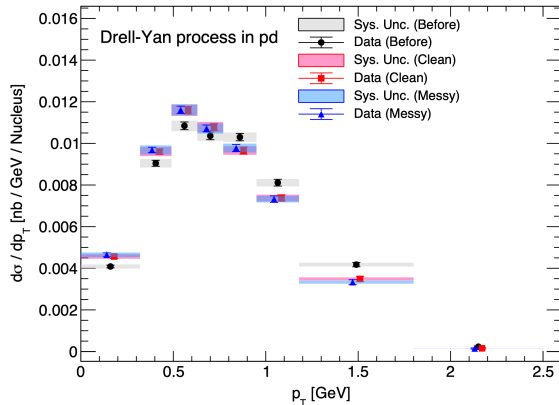
LD2 Target

Unfolded Single Differential Cross-Section as a function of p_T

LH2 Target

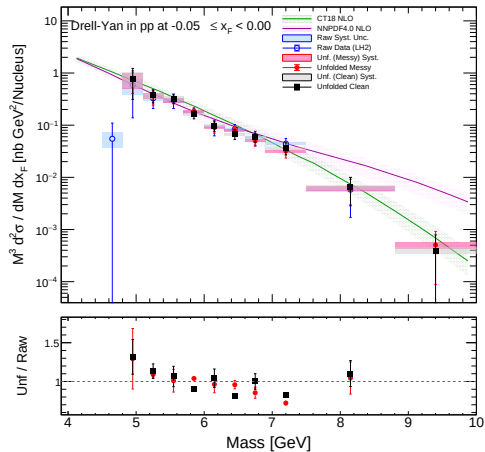


LD2 Target

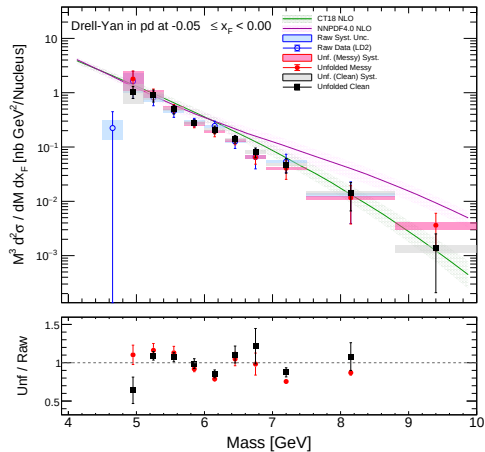


$$-0.05 \leq x_F < 0.00$$

LH2 Target

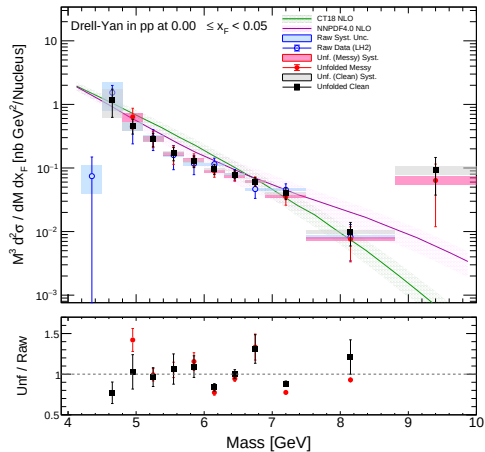


LD2 Target

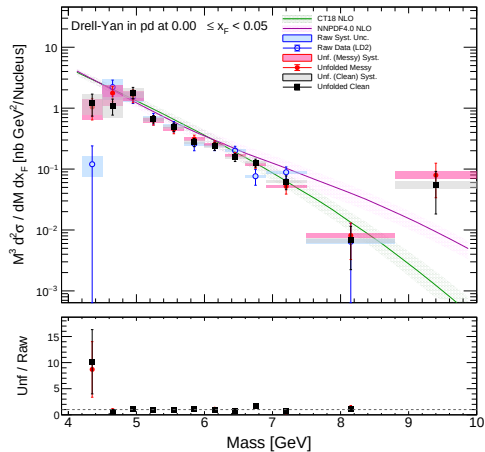


$$0.00 \leq x_F < 0.05$$

LH2 Target

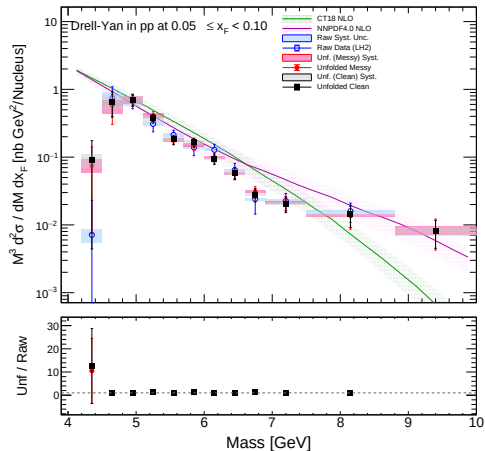


LD2 Target

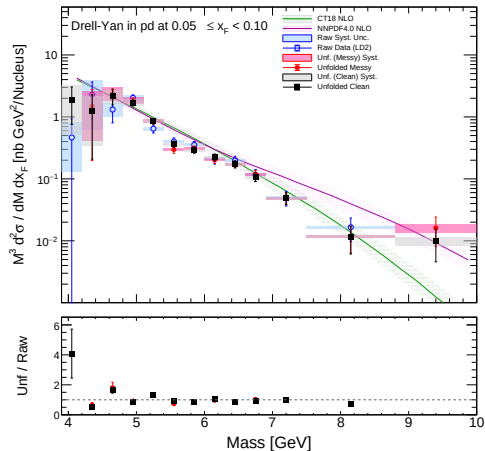


$$0.05 \leq x_F < 0.10$$

LH2 Target

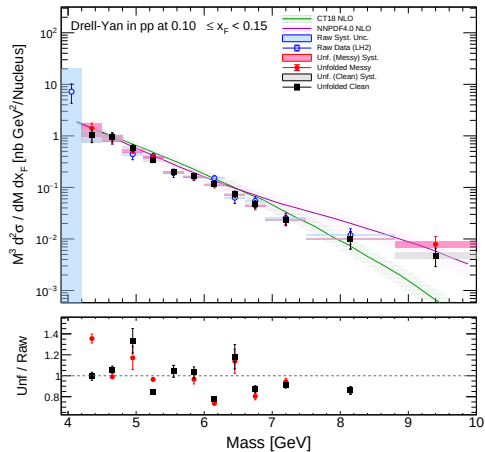


LD2 Target

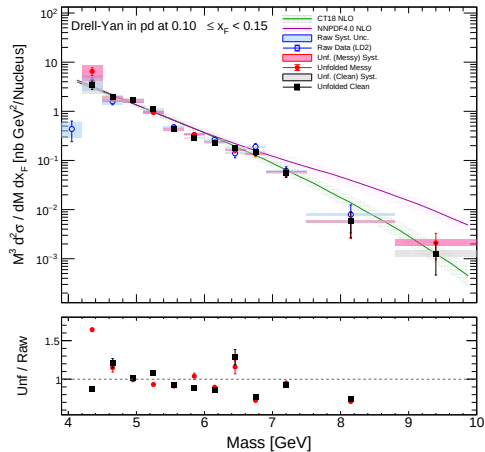


$$0.10 \leq x_F < 0.15$$

LH2 Target

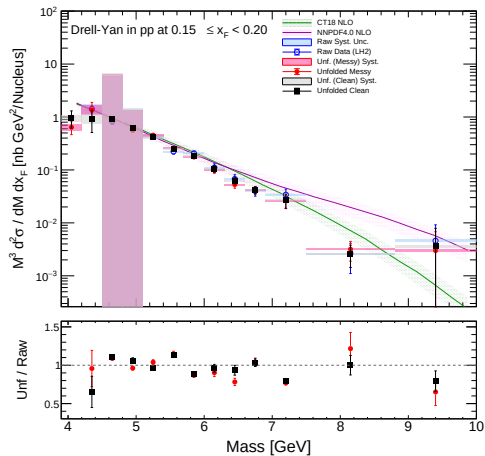


LD2 Target

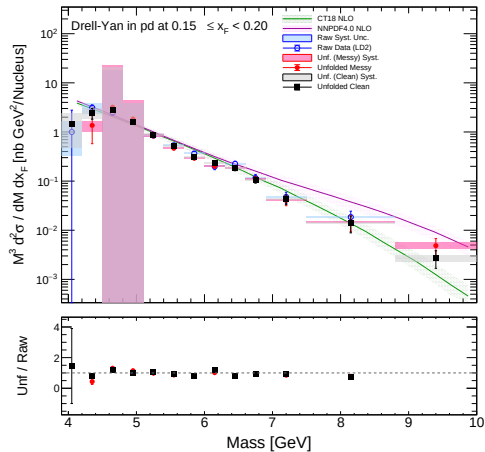


$$0.15 \leq x_F < 0.20$$

LH2 Target

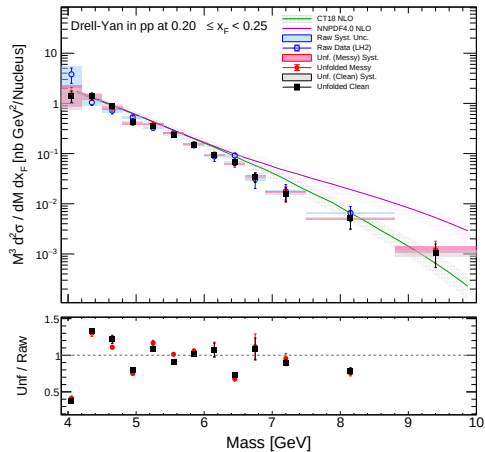


LD2 Target

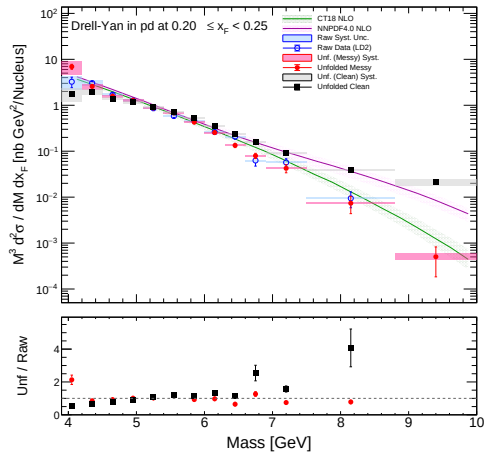


$$0.20 \leq x_F < 0.25$$

LH2 Target

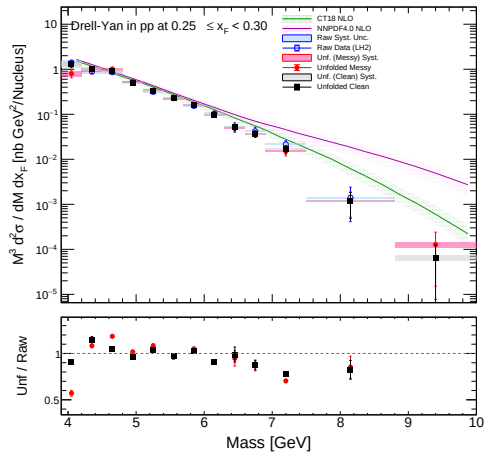


LD2 Target

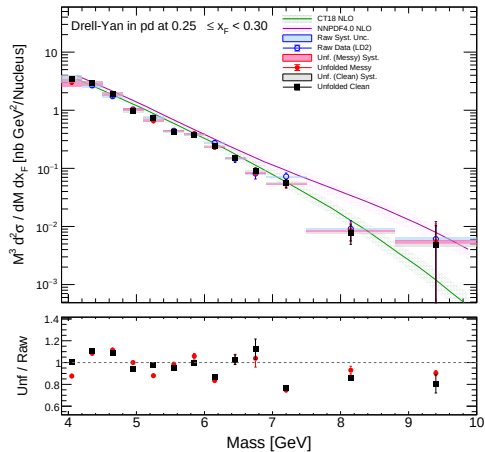


$$0.25 \leq x_F < 0.30$$

LH2 Target

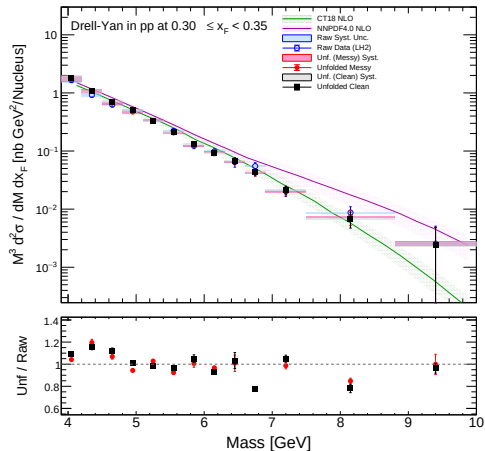


LD2 Target

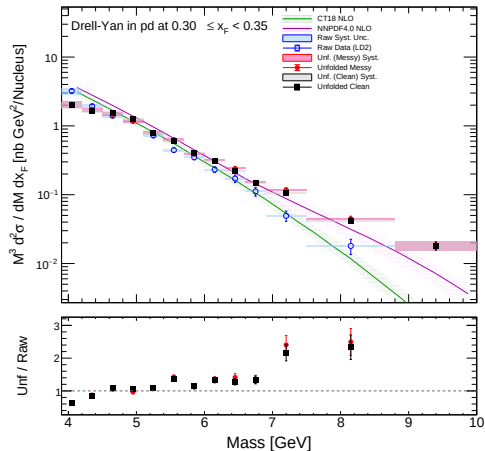


$$0.30 \leq x_F < 0.35$$

LH2 Target

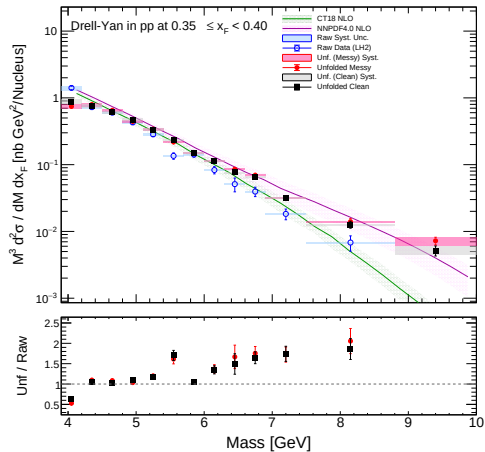


LD2 Target

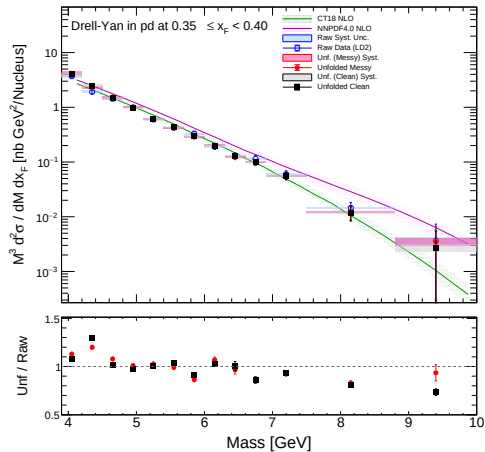


$$0.35 \leq x_F < 0.40$$

LH2 Target

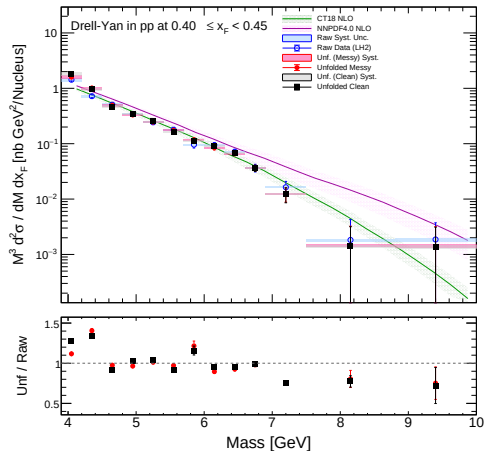


LD2 Target

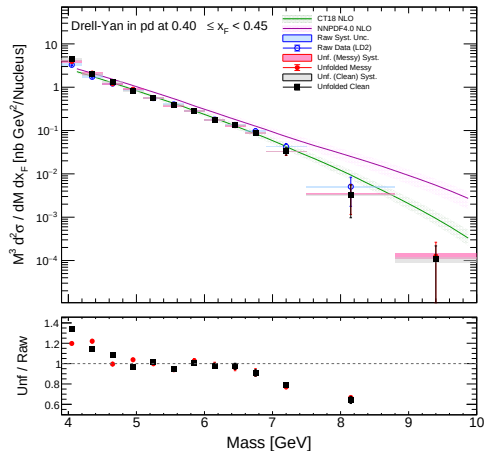


$$0.40 \leq x_F < 0.45$$

LH2 Target

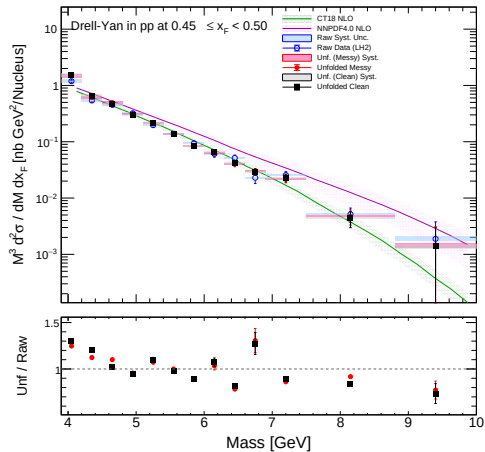


LD2 Target

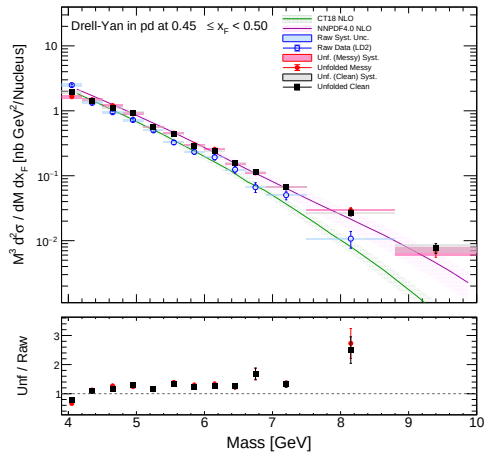


$$0.45 \leq x_F < 0.50$$

LH2 Target

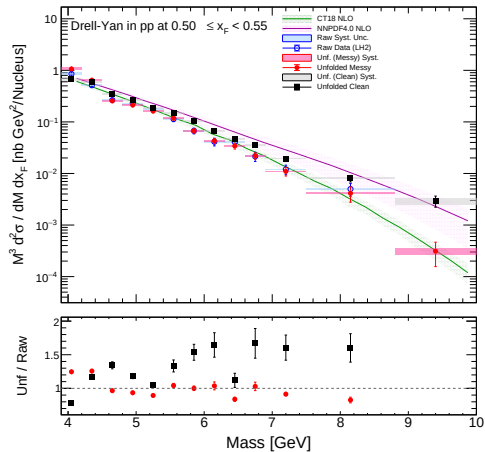


LD2 Target

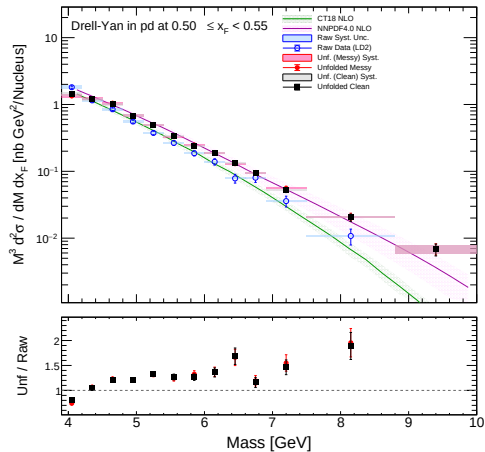


$$0.50 \leq x_F < 0.55$$

LH2 Target

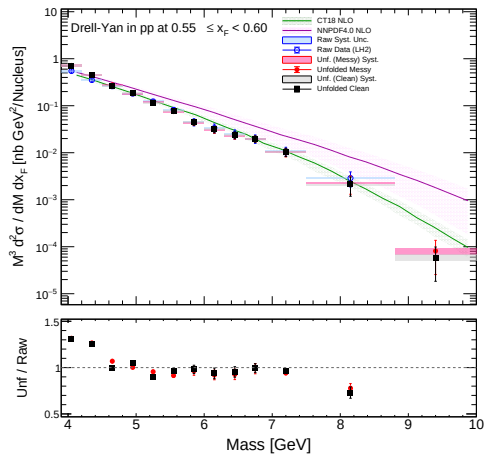


LD2 Target

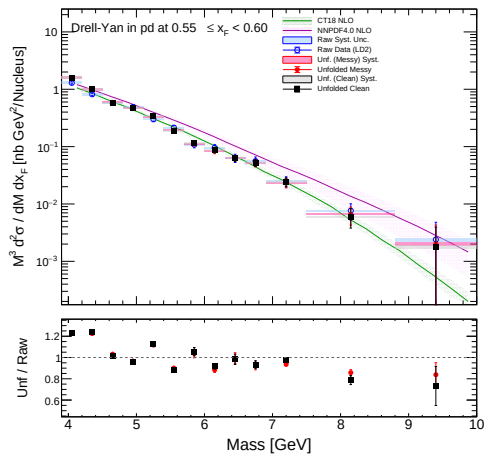


$$0.55 \leq x_F < 0.60$$

LH2 Target

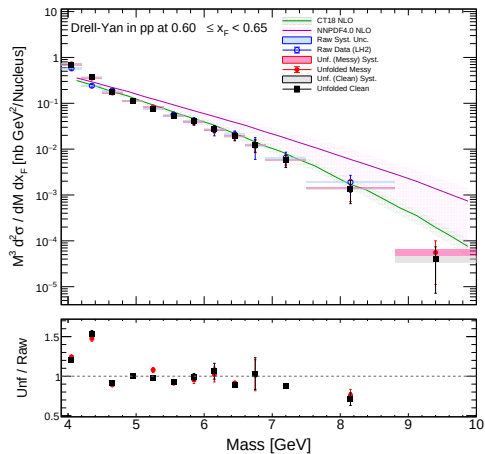


LD2 Target

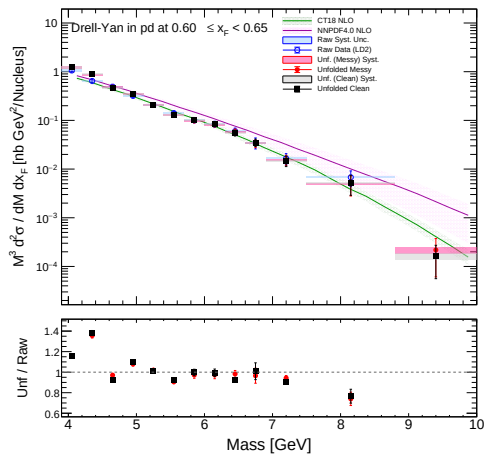


$$0.60 \leq x_F < 0.65$$

LH2 Target

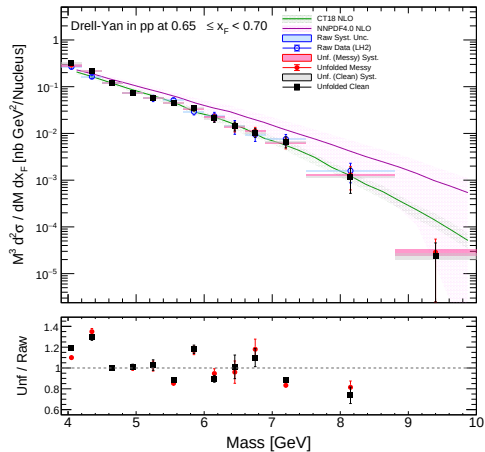


LD2 Target

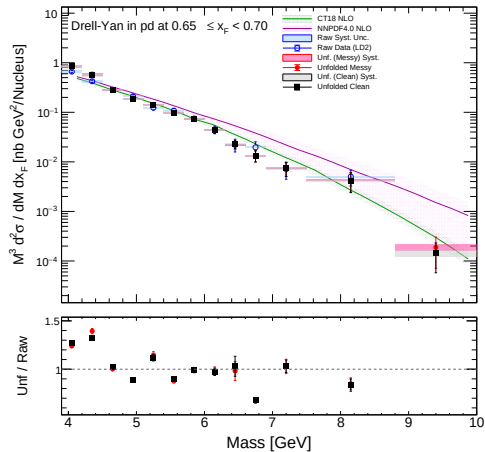


$$0.65 \leq x_F < 0.70$$

LH2 Target

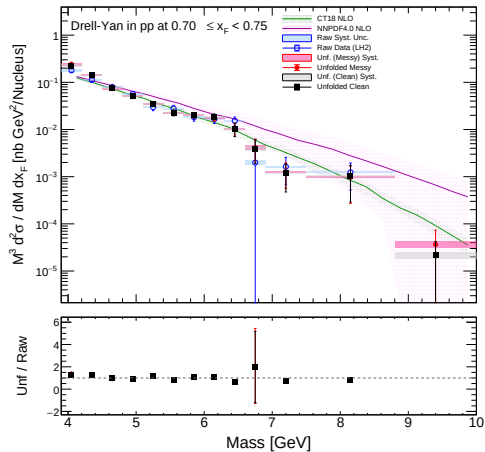


LD2 Target

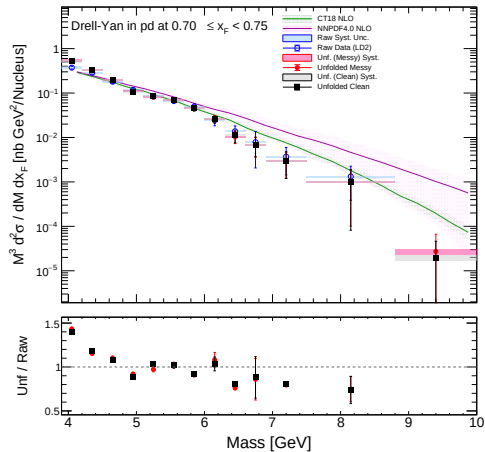


$$0.70 \leq x_F < 0.75$$

LH2 Target

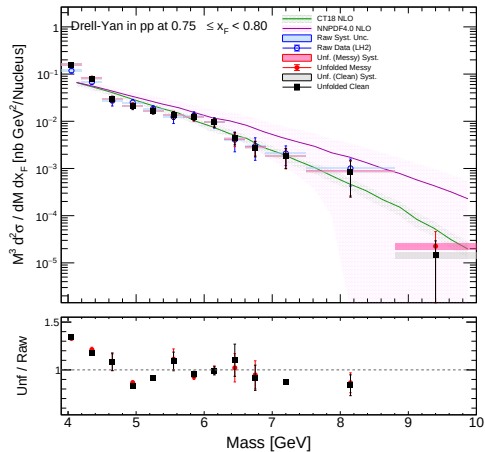


LD2 Target

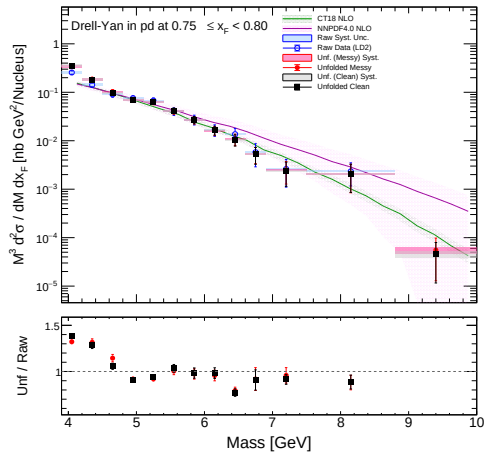


$$0.75 \leq x_F < 0.80$$

LH2 Target

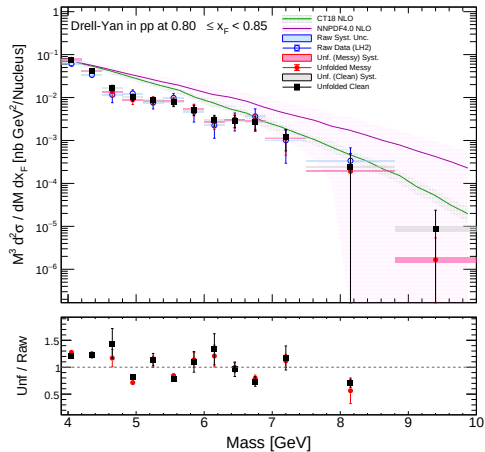


LD2 Target

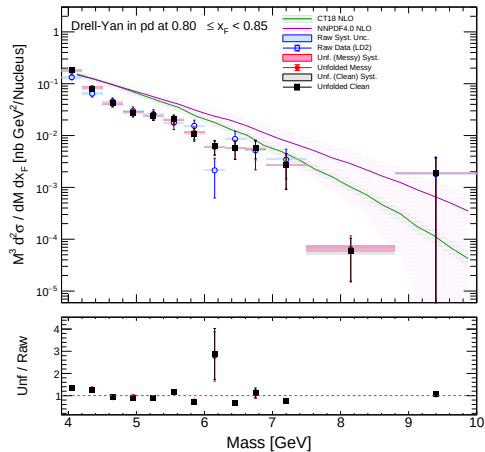


$$0.80 \leq x_F < 0.85$$

LH2 Target



LD2 Target



Conclusions and Next Steps

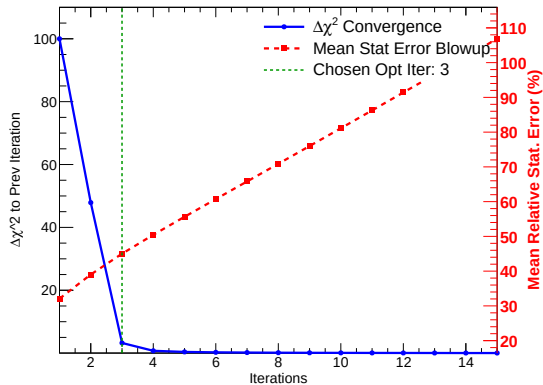
- Unfolding calculation was added for double differential cross-section successfully
- Need to recalculate acceptance correction for Runs: [2, 3] separately (Kenichi and Harsha are working on it)
- Finalizing the draft for the publication and requesting collaboration approval

Backup Slides: Unfolding Optimization

Opt (Messy): $-0.05 \leq x_F < 0.00$

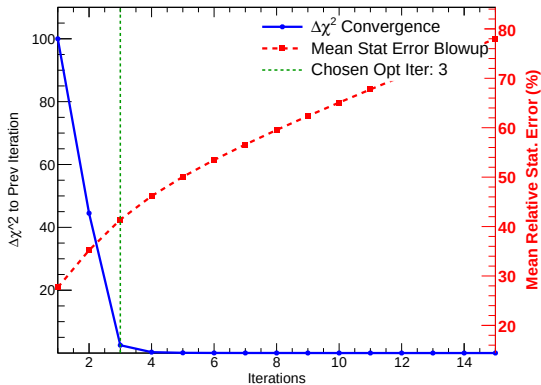
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 0



LD2 (Messy)

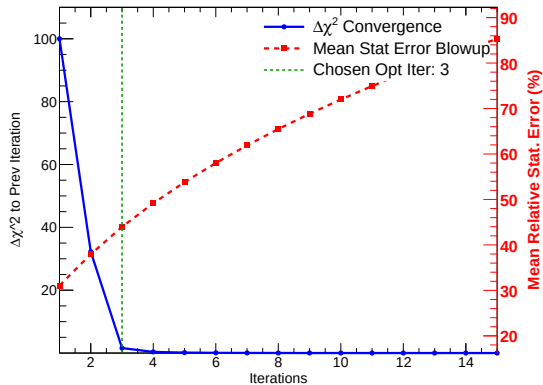
Unfold Optimization LD2 Messy xF Bin 0



Opt (Messy): $0.00 \leq x_F < 0.05$

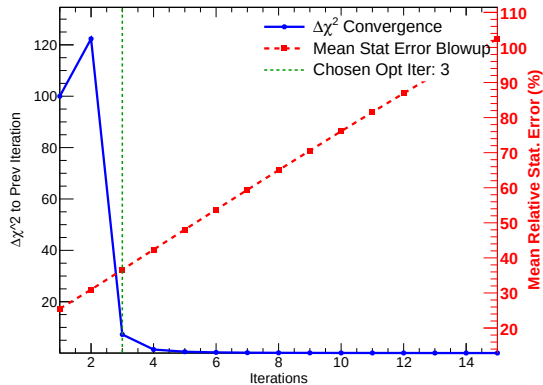
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 1



LD2 (Messy)

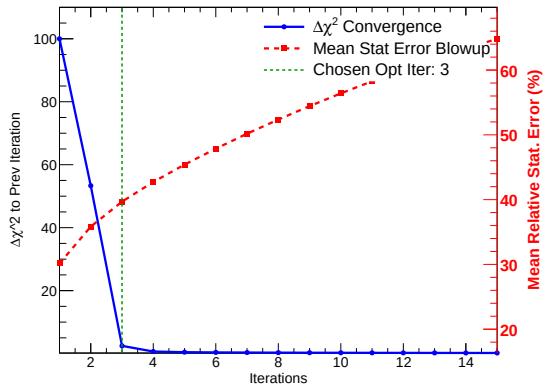
Unfold Optimization LD2 Messy xF Bin 1



Opt (Messy): $0.05 \leq x_F < 0.10$

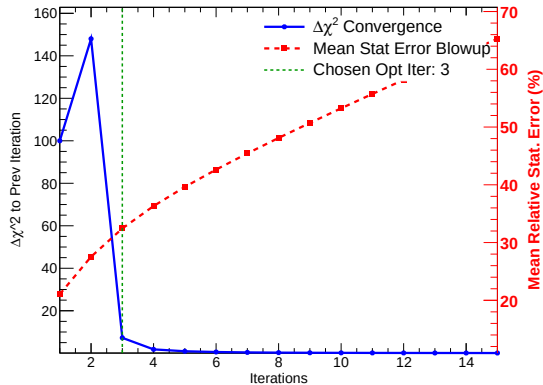
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 2



LD2 (Messy)

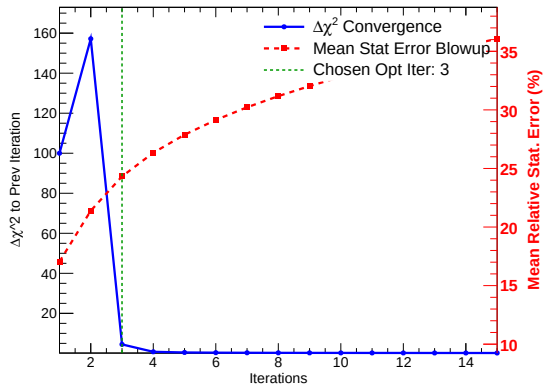
Unfold Optimization LD2 Messy xF Bin 2



Opt (Messy): $0.10 \leq x_F < 0.15$

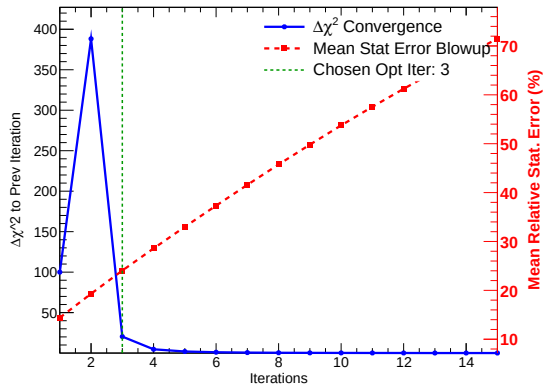
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 3



LD2 (Messy)

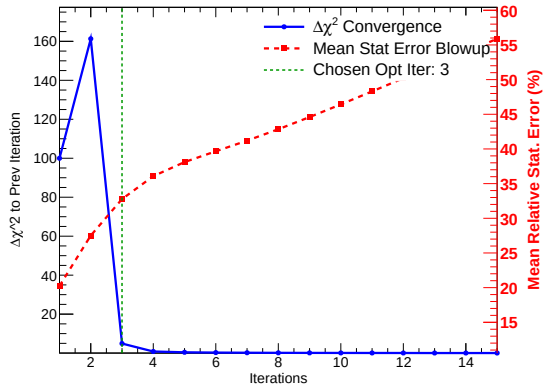
Unfold Optimization LD2 Messy xF Bin 3



Opt (Messy): $0.15 \leq x_F < 0.20$

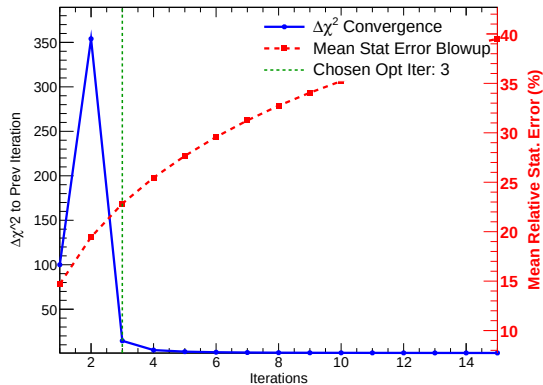
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 4



LD2 (Messy)

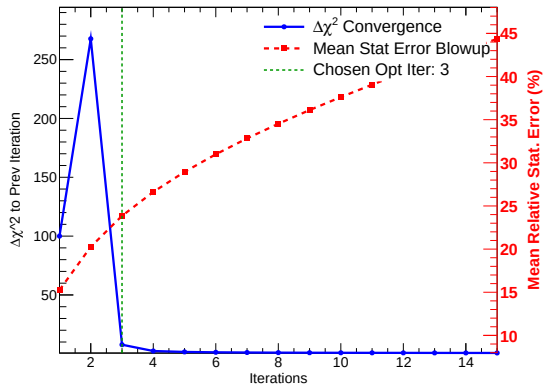
Unfold Optimization LD2 Messy xF Bin 4



Opt (Messy): $0.20 \leq x_F < 0.25$

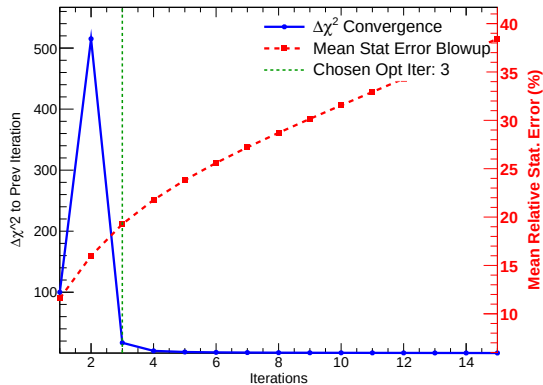
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 5



LD2 (Messy)

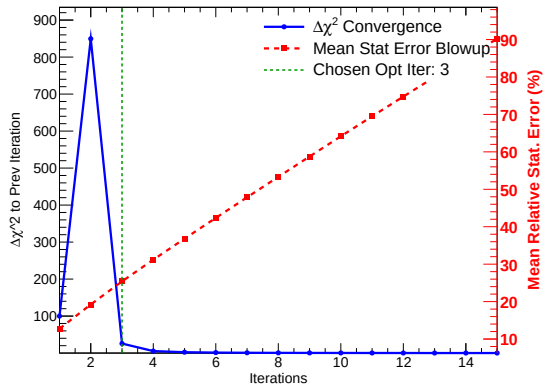
Unfold Optimization LD2 Messy xF Bin 5



Opt (Messy): $0.25 \leq x_F < 0.30$

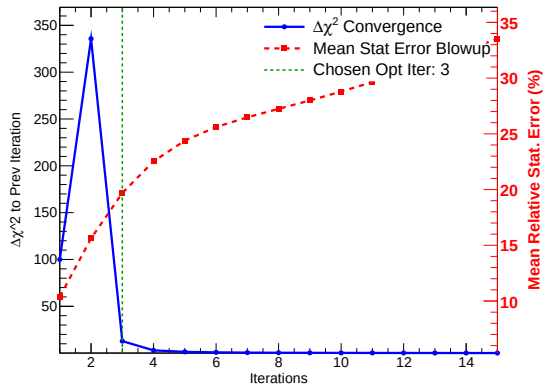
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 6



LD2 (Messy)

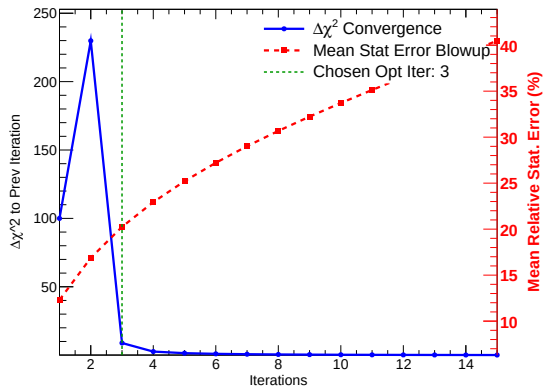
Unfold Optimization LD2 Messy xF Bin 6



Opt (Messy): $0.30 \leq x_F < 0.35$

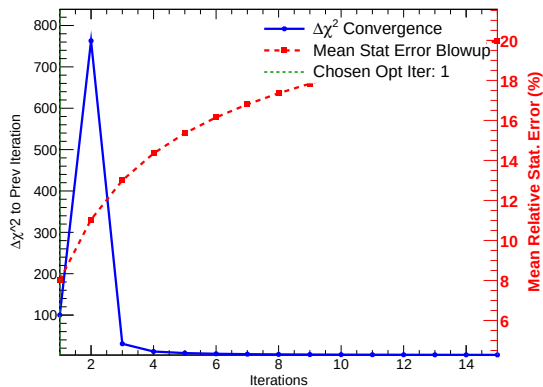
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 7



LD2 (Messy)

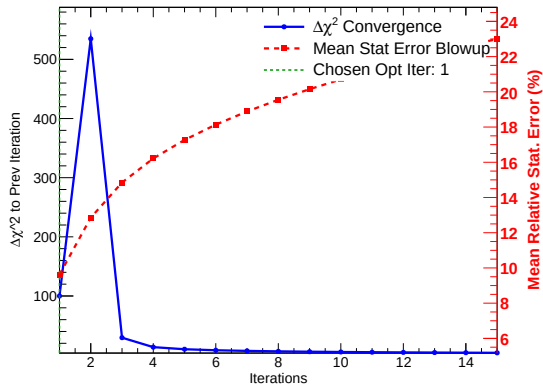
Unfold Optimization LD2 Messy xF Bin 7



Opt (Messy): $0.35 \leq x_F < 0.40$

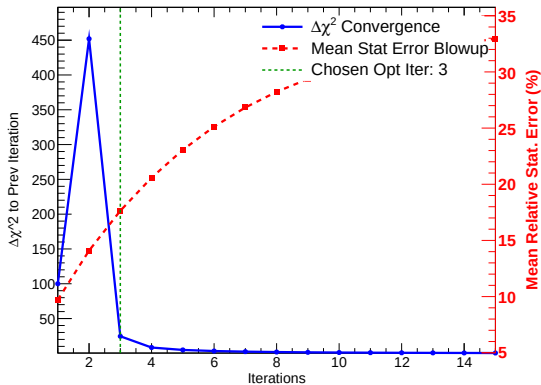
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 8



LD2 (Messy)

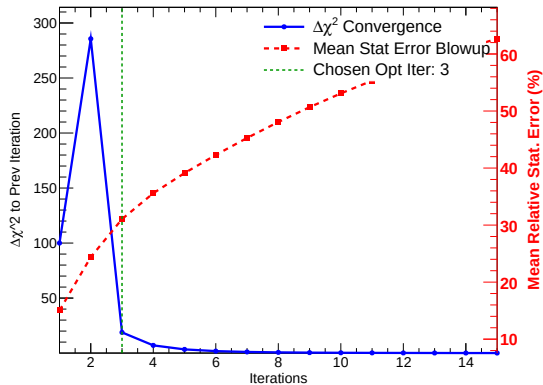
Unfold Optimization LD2 Messy xF Bin 8



Opt (Messy): $0.40 \leq x_F < 0.45$

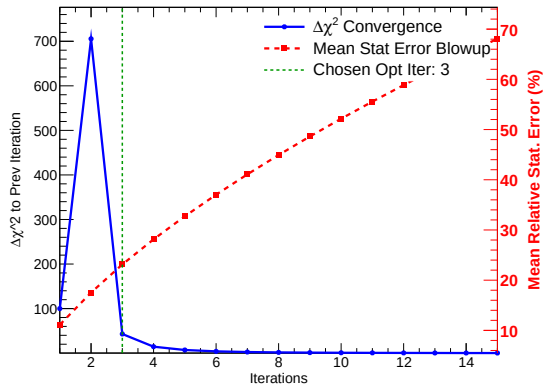
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 9



LD2 (Messy)

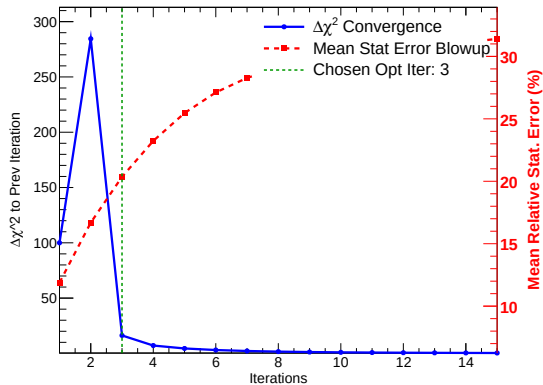
Unfold Optimization LD2 Messy xF Bin 9



Opt (Messy): $0.45 \leq x_F < 0.50$

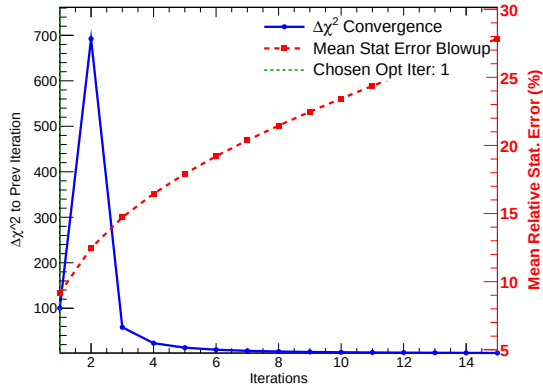
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 10



LD2 (Messy)

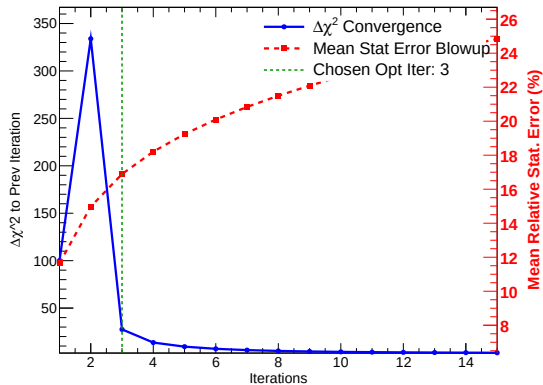
Unfold Optimization LD2 Messy xF Bin 10



Opt (Messy): $0.50 \leq x_F < 0.55$

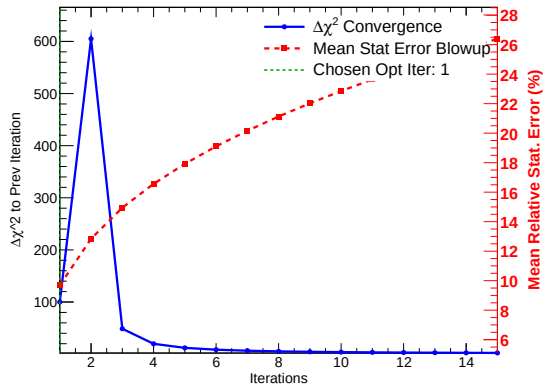
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 11



LD2 (Messy)

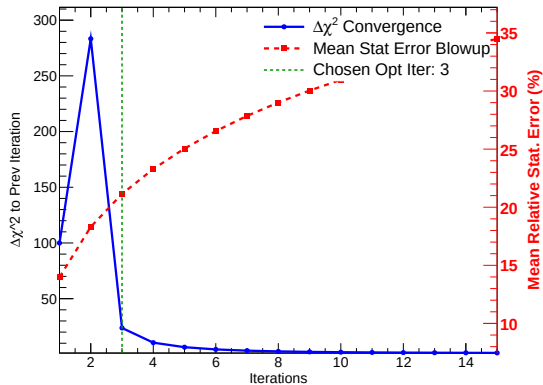
Unfold Optimization LD2 Messy xF Bin 11



Opt (Messy): $0.55 \leq x_F < 0.60$

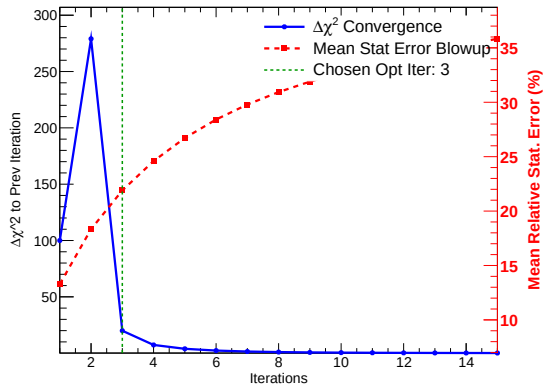
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 12



LD2 (Messy)

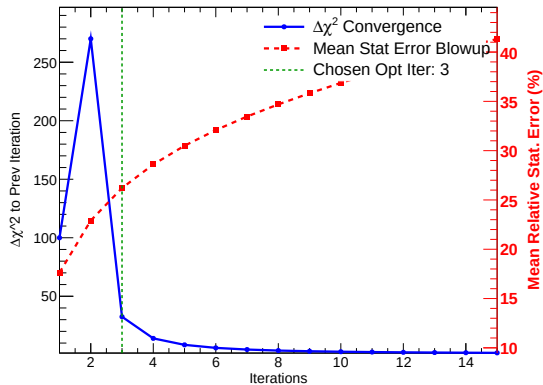
Unfold Optimization LD2 Messy xF Bin 12



Opt (Messy): $0.60 \leq x_F < 0.65$

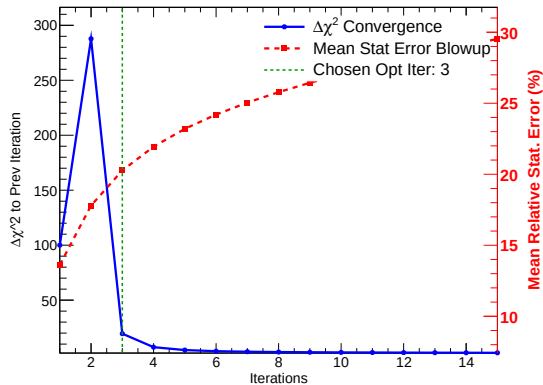
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 13



LD2 (Messy)

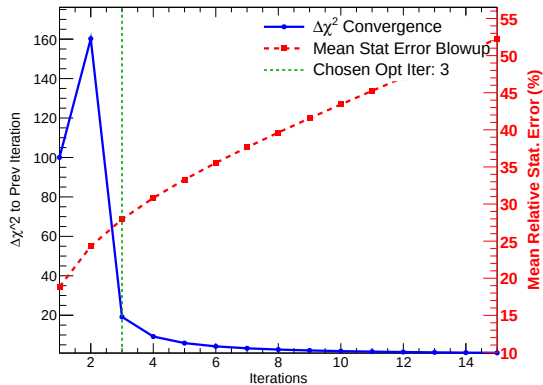
Unfold Optimization LD2 Messy xF Bin 13



Opt (Messy): $0.65 \leq x_F < 0.70$

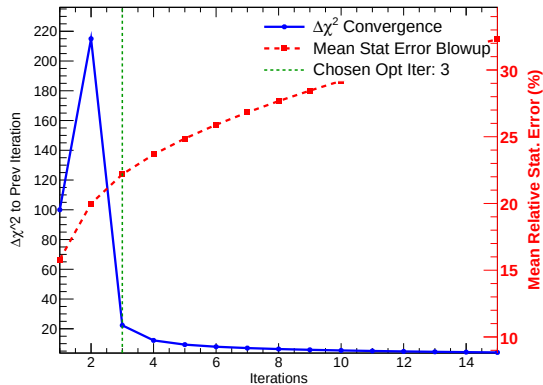
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 14



LD2 (Messy)

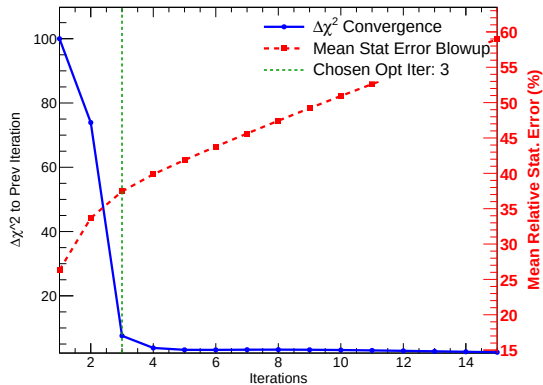
Unfold Optimization LD2 Messy xF Bin 14



Opt (Messy): $0.70 \leq x_F < 0.75$

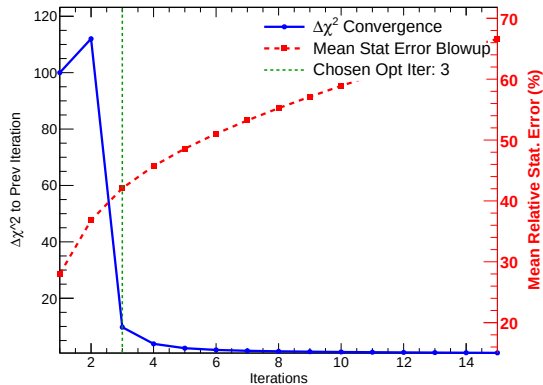
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 15



LD2 (Messy)

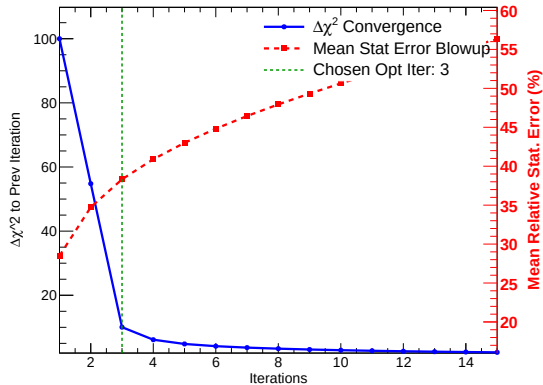
Unfold Optimization LD2 Messy xF Bin 15



Opt (Messy): $0.75 \leq x_F < 0.80$

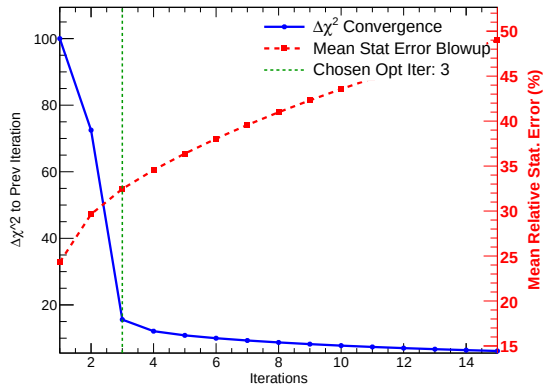
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 16



LD2 (Messy)

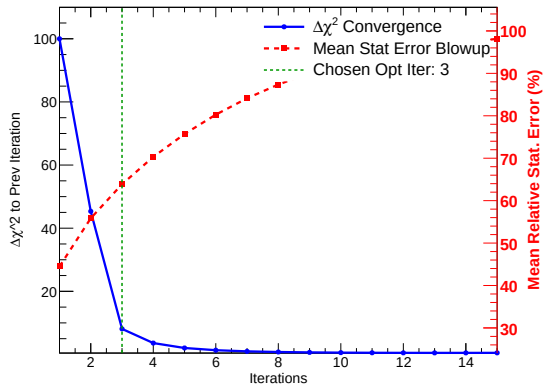
Unfold Optimization LD2 Messy xF Bin 16



Opt (Messy): $0.80 \leq x_F < 0.85$

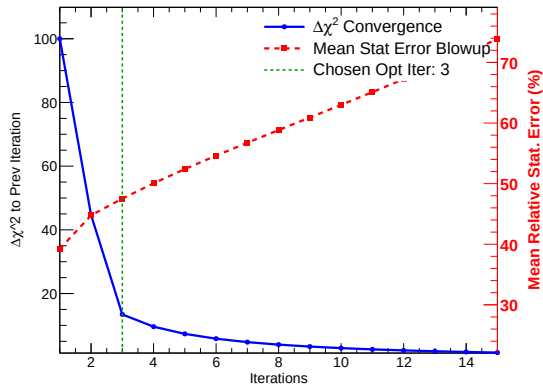
LH2 (Messy)

Unfold Optimization LH2 Messy xF Bin 17



LD2 (Messy)

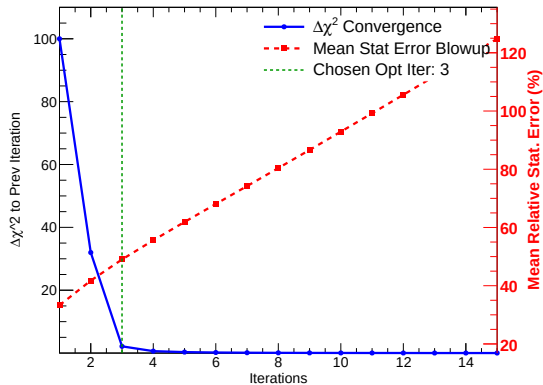
Unfold Optimization LD2 Messy xF Bin 17



Opt (Clean): $-0.05 \leq x_F < 0.00$

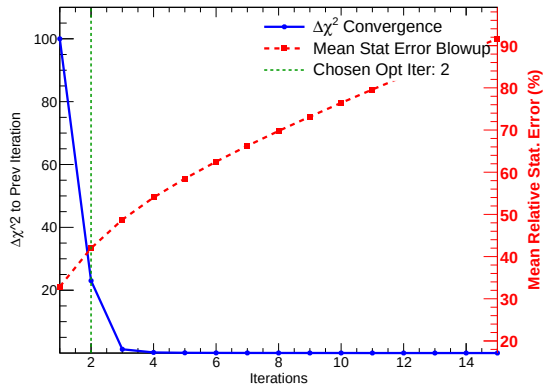
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 0



LD2 (Clean)

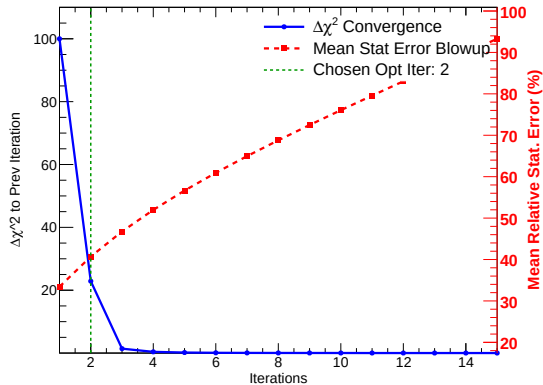
Unfold Optimization LD2 Clean xF Bin 0



Opt (Clean): $0.00 \leq x_F < 0.05$

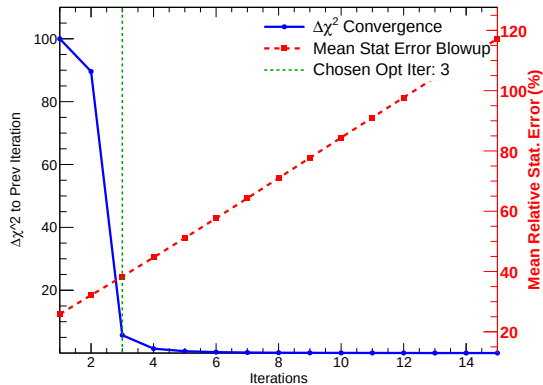
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 1



LD2 (Clean)

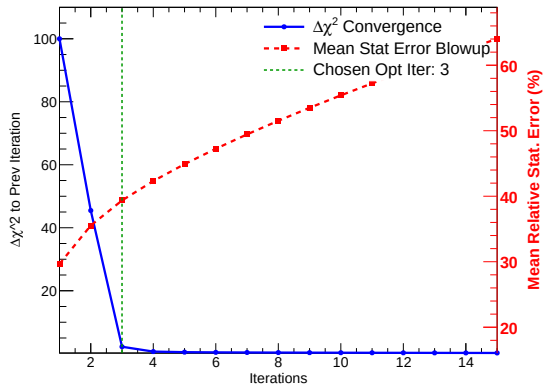
Unfold Optimization LD2 Clean xF Bin 1



Opt (Clean): $0.05 \leq x_F < 0.10$

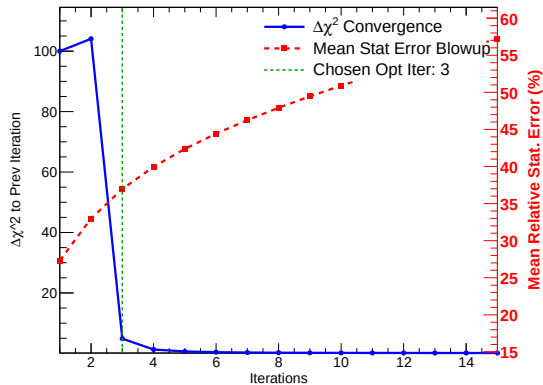
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 2



LD2 (Clean)

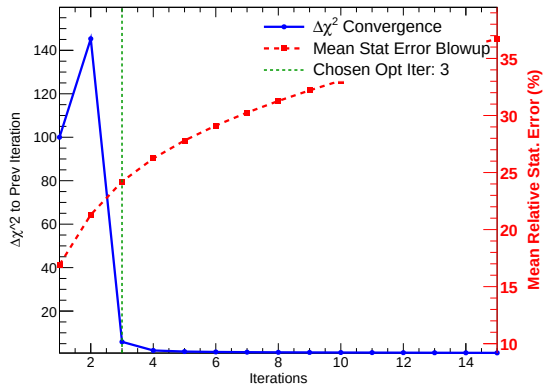
Unfold Optimization LD2 Clean xF Bin 2



Opt (Clean): $0.10 \leq x_F < 0.15$

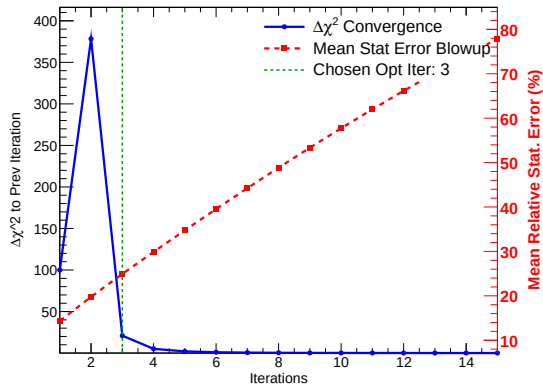
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 3



LD2 (Clean)

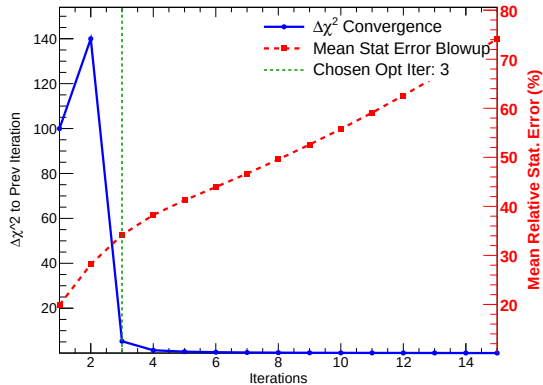
Unfold Optimization LD2 Clean xF Bin 3



Opt (Clean): $0.15 \leq x_F < 0.20$

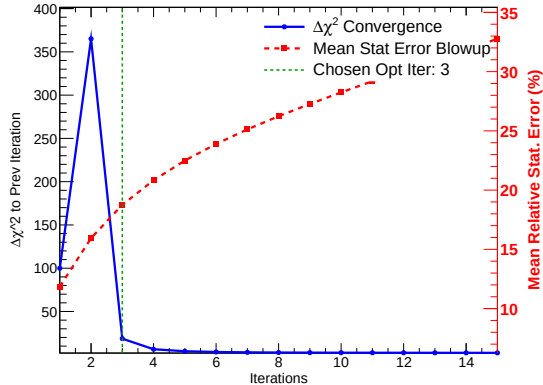
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 4



LD2 (Clean)

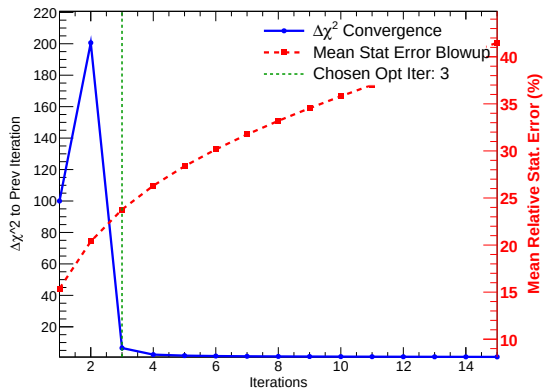
Unfold Optimization LD2 Clean xF Bin 4



Opt (Clean): $0.20 \leq x_F < 0.25$

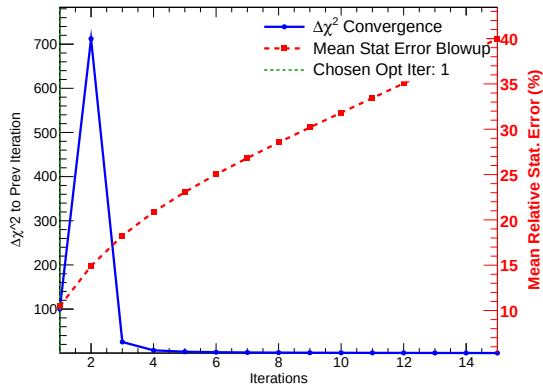
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 5



LD2 (Clean)

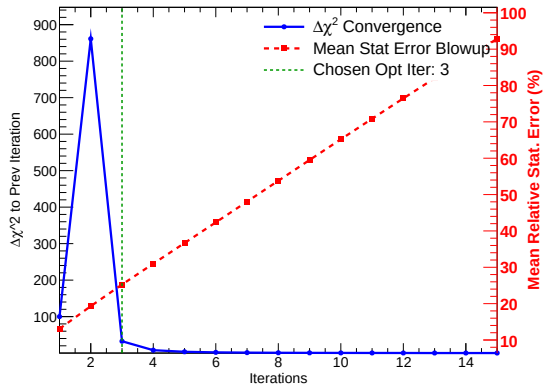
Unfold Optimization LD2 Clean xF Bin 5



Opt (Clean): $0.25 \leq x_F < 0.30$

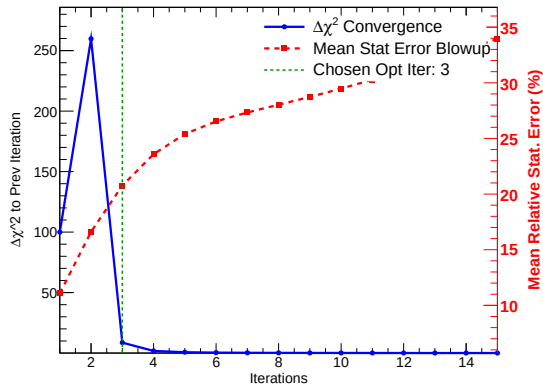
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 6



LD2 (Clean)

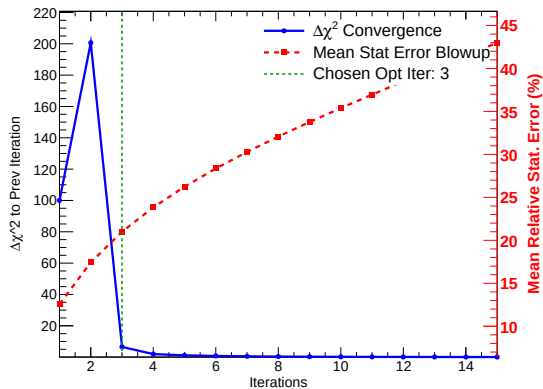
Unfold Optimization LD2 Clean xF Bin 6



Opt (Clean): $0.30 \leq x_F < 0.35$

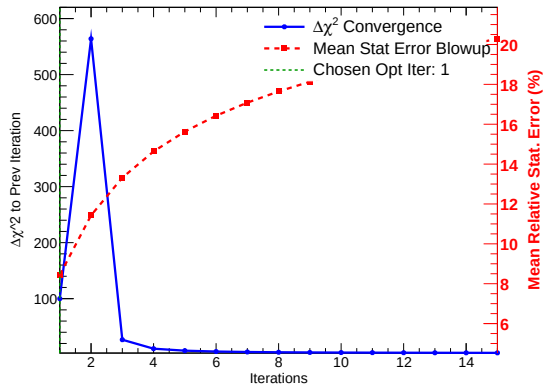
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 7



LD2 (Clean)

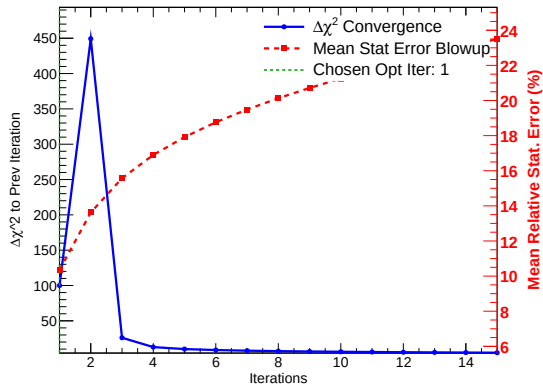
Unfold Optimization LD2 Clean xF Bin 7



Opt (Clean): $0.35 \leq x_F < 0.40$

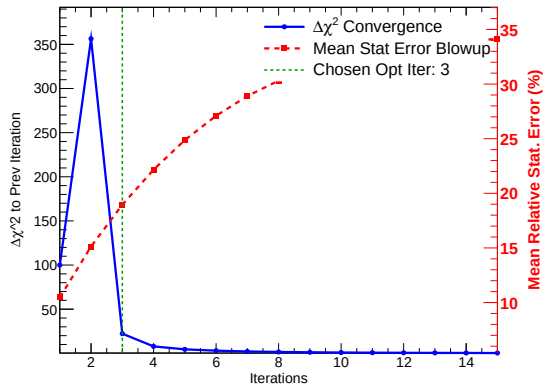
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 8



LD2 (Clean)

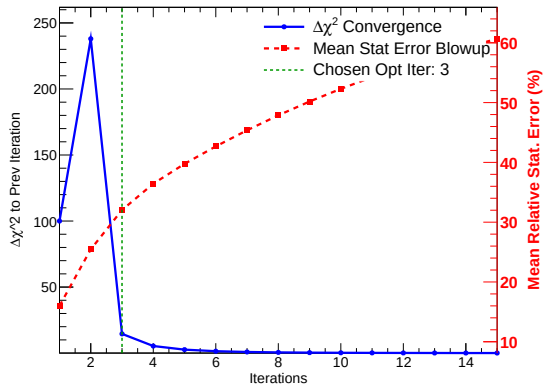
Unfold Optimization LD2 Clean xF Bin 8



Opt (Clean): $0.40 \leq x_F < 0.45$

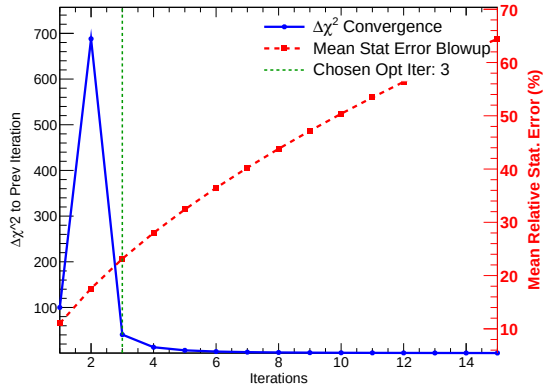
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 9



LD2 (Clean)

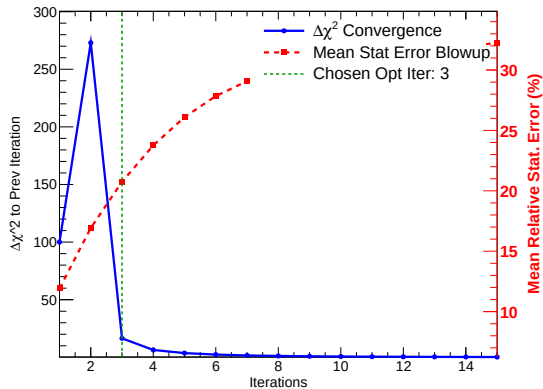
Unfold Optimization LD2 Clean xF Bin 9



Opt (Clean): $0.45 \leq x_F < 0.50$

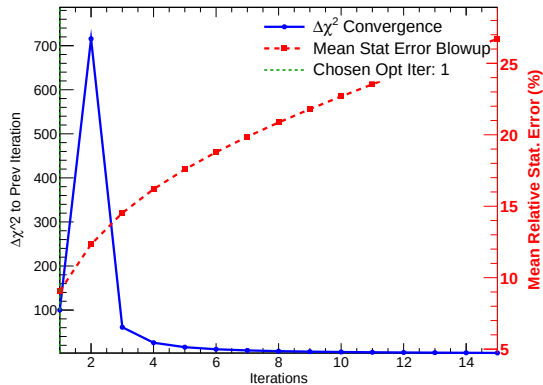
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 10



LD2 (Clean)

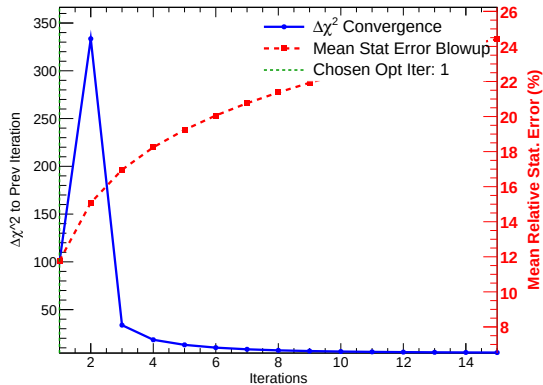
Unfold Optimization LD2 Clean xF Bin 10



Opt (Clean): $0.50 \leq x_F < 0.55$

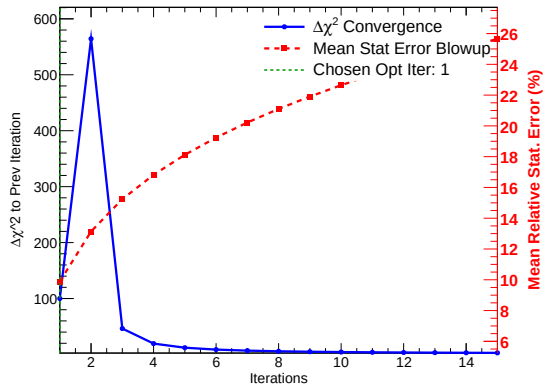
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 11



LD2 (Clean)

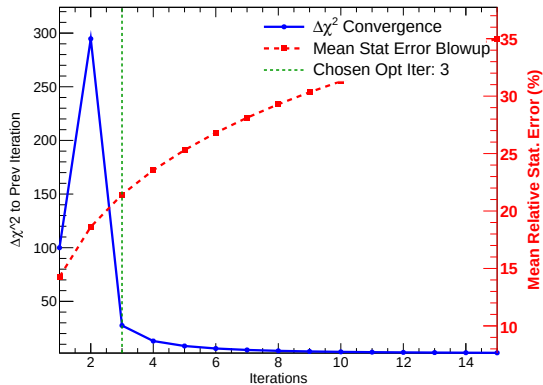
Unfold Optimization LD2 Clean xF Bin 11



Opt (Clean): $0.55 \leq x_F < 0.60$

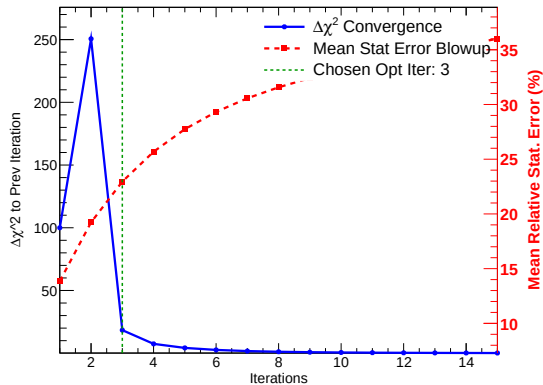
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 12



LD2 (Clean)

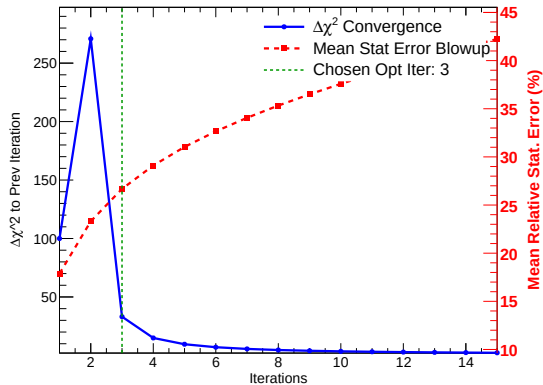
Unfold Optimization LD2 Clean xF Bin 12



Opt (Clean): $0.60 \leq x_F < 0.65$

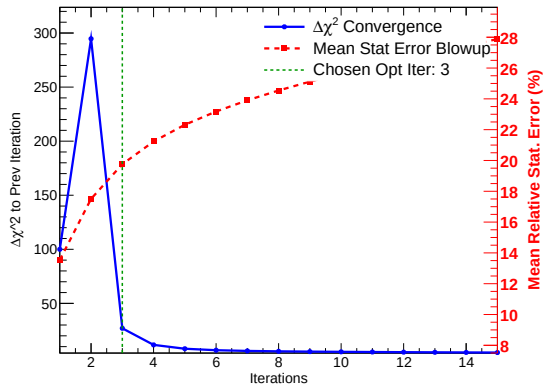
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 13



LD2 (Clean)

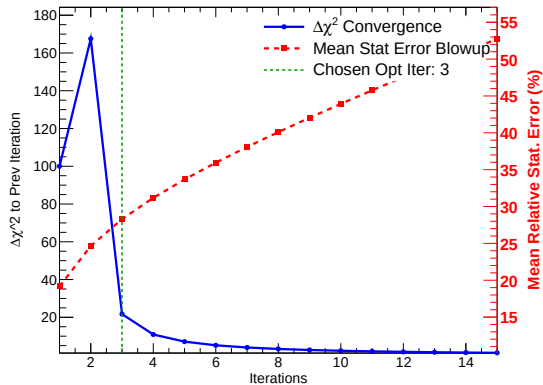
Unfold Optimization LD2 Clean xF Bin 13



Opt (Clean): $0.65 \leq x_F < 0.70$

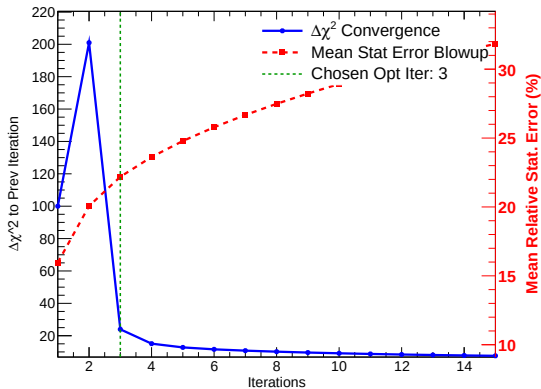
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 14



LD2 (Clean)

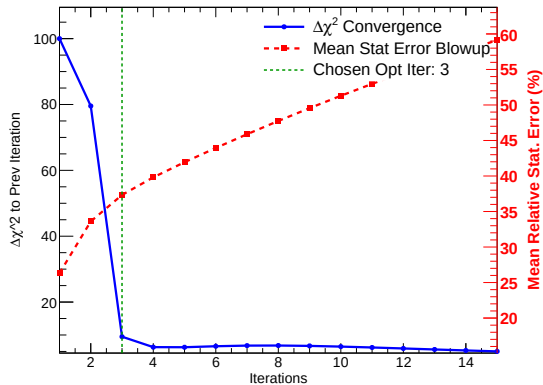
Unfold Optimization LD2 Clean xF Bin 14



Opt (Clean): $0.70 \leq x_F < 0.75$

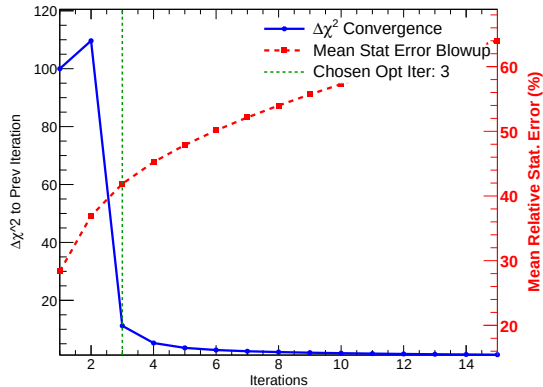
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 15



LD2 (Clean)

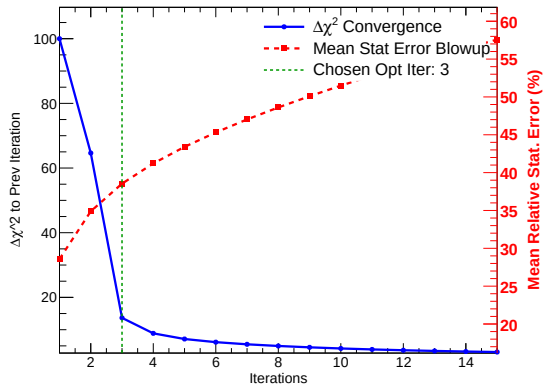
Unfold Optimization LD2 Clean xF Bin 15



Opt (Clean): $0.75 \leq x_F < 0.80$

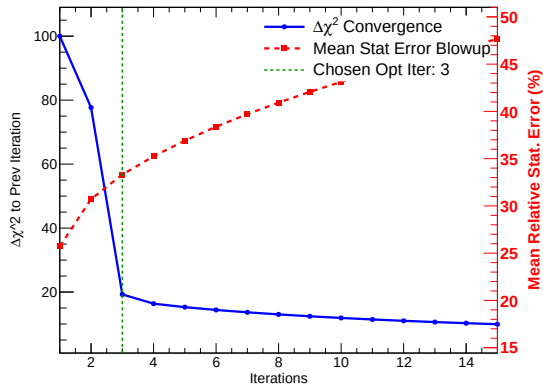
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 16



LD2 (Clean)

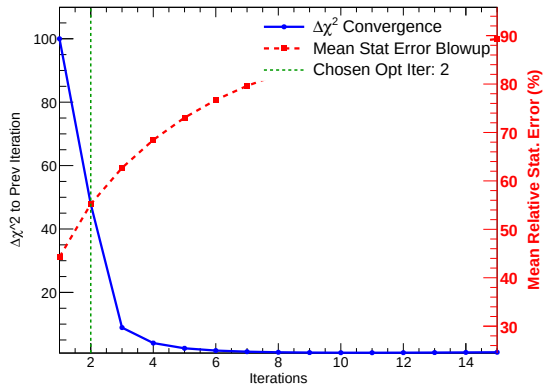
Unfold Optimization LD2 Clean xF Bin 16



Opt (Clean): $0.80 \leq x_F < 0.85$

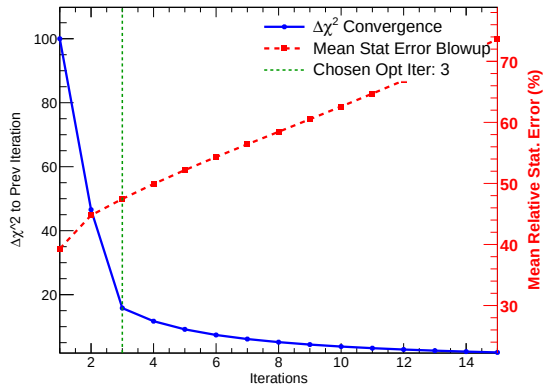
LH2 (Clean)

Unfold Optimization LH2 Clean xF Bin 17



LD2 (Clean)

Unfold Optimization LD2 Clean xF Bin 17



Delta χ^2 determination

In iterative Bayesian unfolding, $\Delta\chi^2$ is used as a convergence metric to measure the change in the unfolded result between iterations.

Convergence Metric Equation

$$\Delta\chi^2 = \sum_{k=1}^{N_{\text{bins}}} \frac{(Y_i(k) - Y_{i-1}(k))^2}{\sigma_i(k)^2}$$

Definition of Variables:

- $Y_i(k)$: Unfolded yield in bin k at the current iteration i .
- $Y_{i-1}(k)$: Unfolded yield in bin k at the previous iteration $i - 1$.
- $\sigma_i(k)$: Statistical uncertainty (error) of bin k at the current iteration i .

The unfolding is considered converged when $\Delta\chi^2 \approx 1.0$, indicating changes are within statistical noise.